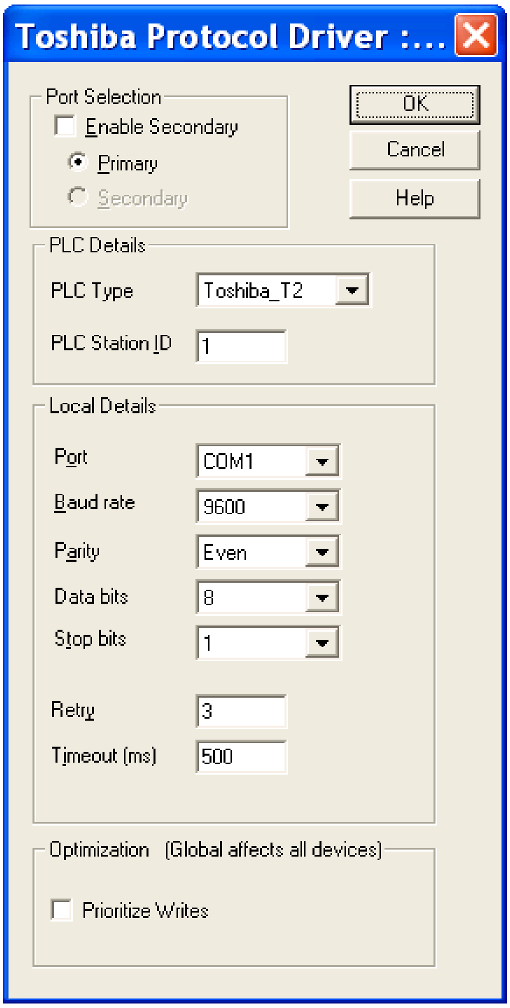
Toshiba ASCII protocol driver

Overview

The TOSHIBA driver currently only supports the ascii-based protocol, the binary protocol is not supported in this driver. This driver supports the PROSEC T2 PLC, EX100 , EX500. Communication is RS-485. The ADROIT TOSHIBA driver will act as the master in a master-slave transaction system.

See also: Toshiba binary driver and Toshiba Ethernet driver for T2.

Device Configuration



PLC Details

PLC Station ID - Up to a maximum of 32 PLCs may be addressed.

Local Details

Port - The available serial ports on your machine.

Baud rate - This is configurable. The TOSHIBA PROSEC T2 PLC supports the following baud rates : 300, 600, 1200, 2400, 4800, 9600 or 19200 bps. Select the appropriate baud rate from the dialogue's drop-down list.

Parity - This is configurable. Valid choices are None, Odd or Even.

Data bits - This is configurable. Valid choices are 7 or 8 bits.

Stop bits - This is configurable. Valid choices are 1 or 2 bits.

Retry - This is the number of times the ADROIT TOSHIBA driver will attempt a particular scan job before failing.

Timeout - This is the maximum time the ADROIT TOSHIBA driver will wait for a response to its request message from the PLC.

Optimisation (Global affects all devices)

Prioritize write – Check this check box to ensure the driver will optimize writes.

Wiring

The Computer Link Port on the PLC must be used for communications between ADROIT and the PLC. This port accepts a 4-wire RS-485 interface. The Computer Link Port is a standard 15-pin female connector. The following table shows the pin assignment of the Computer Link Port :

Pin No.	Signal name	Description	Signal direction
1	**FG**	**Protective ground**	**PLC ↔ Host**
7	**SG**	**Signal ground**	**PLC ↔ Host**
8	**SG**	"	**PLC ↔ Host**
15	**SG**	"	**PLC ↔ Host**
3	**TXA**	**Transmitted data**	**PLC → Host**
11	**TXB**	"	**PLC → Host**
2	**RXA**	**Received data**	**PLC ← Host**
10	**RXB**	"	**PLC ← Host**
5	**RTSA**	**Request to send**	**PLC → Host**
13	**RTSB**	"	**PLC → Host**
4	**CTSA**	**Clear to send**	**PLC ← Host**
12	**CTSB**	"	**PLC ← Host**

The driver was tested with the PC labs RS 422/485 card.

Supported Commands

Area (Register/Device)	Format (Integer/Real)	Format (Boolean)	Format (Strings)	Description
XW	XW0	XW0.0 - XW0.15	N/A	External Input Register
YW	YW0	YW0.0 - YW0.15	N/A	External Output Register
ZW	ZW0	ZW0.0 - ZW0.15	N/A	Link Relay Register
RW	RW0	RW0.0 - RW0.15	N/A	Auxiliary Relay Register
D	D0	D0.0 - D0.15	D0 S2	Data Register
C	C0	C0.0 - C0.15	N/A	Counter Register
T	T0	T0.0 - T0.15	N/A	Timer Register

Note: Strings are only supported for Data Registers by Adding “S” at the end of the address with the amount of characters required!

Example: D:100 S20

Message protocol

The checksum value is the ASCII code of the lower two digits of the sum of all the values of all the bytes in a message starting from the first byte until the “&” termination byte.

Telegram examples shown below. All values shown are in ASCII-HEX unless specified otherwise.

Read request:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Start Register / Device variable bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	DR	XW0	&	<<>>	)	CR

Read response:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Data variable bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	DR	<<>>	&	<<>>	)	CR

Write request:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Start Register / Device variable bytes	Data Delimiter	Number of data bytes	Data variable bytes
(	A	01	DR	XW0	,	1	<<>>

… continued …

Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
&	<<>>	)	CR

Acknowledgment:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Status 4 bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	ST	0002	&	<<>>	)	CR

Advanced Registry key settings

The following settings are not configurable in the normal fashion and are accessible via the registry.

Open the registry editor and go to …
HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\TOSHIBA

..then open each device, and open "Primary". Add the appropriate key (DWORD)

> <u>Close port on failure:</u> If ethernet serial extenders hang, try changing the value ***ClosePortOnFailure***, If the key does not exist add it as type dword , 1 means close port if failed , 0 means don't close the port. Driver default is 1.

Document Revisions

Date	Changes
11-Apr-05	Fixed document margins and minor formatting adjustments done. Some redundant statements removed.
16-Apr-08	Added String support to Driver on Data Registers. – Rev 13